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CE F230
Civil Engineering Materials

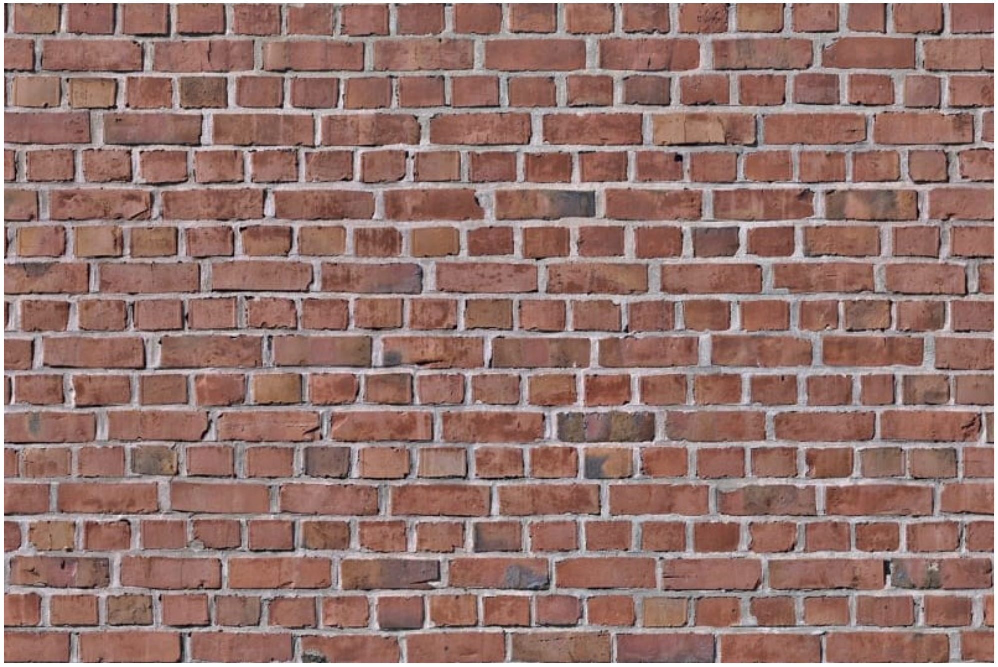
Clay and clay products

BITS Pilani
Pilani Campus

Masonry

Masonry

- A masonry structure is an assemblage of masonry units and mortar



- **Masonry units:**
  - ✓ Stone
  - ✓ Clay bricks
  - ✓ Sand lime bricks
  - ✓ Fly ash bricks
  - ✓ Concrete blocks
  - ✓ Autoclave cellular concrete

Masonry

- **Various applications:**
  - ✓ Building walls
  - ✓ Fireplaces and chimneys
  - ✓ Garden walls and fences
  - ✓ Retaining walls
  - ✓ Pavement
  - ✓ Foundation blocks

Clay

- Clay is an earthen mineral mass or fragmentary rock
- It is capable of mixing with water and forming plastic viscous mass
- Clay retains its shape when moulded and dried
  - When such masses are heated to redness, they acquire hardness and strength
- Purest clays consist mainly of kaolinite with small quantities of minerals such as quartz, mica, felspar, magnesite

Classification of Clay

On basis of Origin

- Residual clays (known as kaolin or china clay)
- Transported or sedimentary clays

- **Residual Clay**

  - Formed from the *decay of underlying rocks*.
  - It is used for making pottery. Eg - Kaoline

- **Transported Clay**

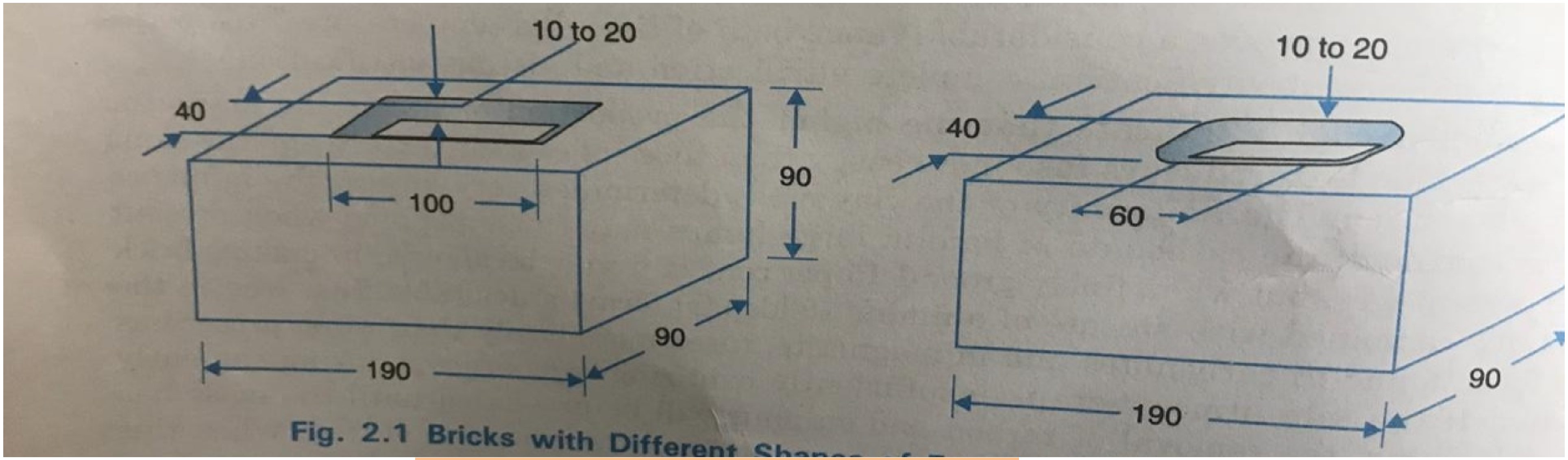
  - Formed due to weathering agencies.

Clay Brick

- One of the oldest building material. *Cheap, durable, easy to handle and deal with.*
- Bricks are rectangular blocks .
- Density of bricks ranges from $2.5 - 2.8\ g/cm^3$
- Modulus of elasticity ranges from $5\text{-}30 \times 10^3\ N/mm^2$
- Clay bricks are *economical* and *easily available*.
- Generally *resistant to alkalis, acids, chemicals* and are excellent *fire resistant* materials

Clay Brick

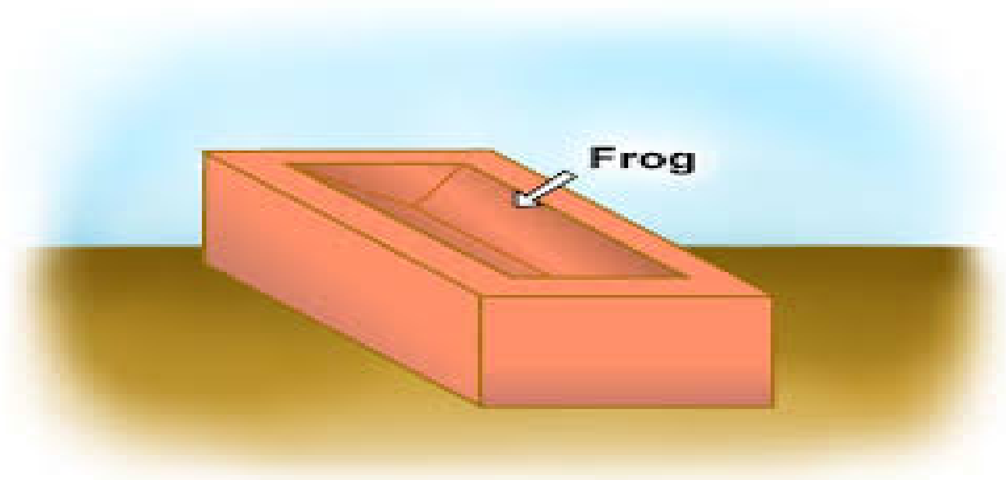
- Size of a standard brick (modular brick) should be 19 x 9 x 9 cm and 19 x 9 x 4 cm
- However, in India, 9 x 4.5 x 3 inch bricks, known as field bricks are mostly available
- Length of brick= 2 x width of brick + thickness of mortar
- Height of brick = width of brick



Bricks with different shapes of frogs

Frog

- An indent of 1-2 cm deep in a brick is known as frog
- Frog forms a key for holding the mortar. Thus, bricks are laid with frogs on top
- The size of frog mostly is 10 x 4 x 1 cm
- It is not provided in 4 cm high bricks



Ingredients and functions of good brick earth

1. Silica 50-60%
   - Enables bricks to retain its shape, imparts durability, prevents shrinkage and warping.
   - Excess silica makes brick brittle and weak on burning.

2. Alumina 20-30%
   - Absorbs water and renders the clay plastic.
   - Excess causes cracks in brick on drying.
   - Clays with exceedingly high alumina content – likely to be very refractory

3. Lime 10%
   - Reduces the shrinkage on drying
   - Causes silica in clay to melt on burning and thus helps to bind it
   - Excess of lime causes the brick to melt and loose its shape.

Ingredients and functions of good brick earth

4. Magnesia- < 1%
   - Reduces warping.
   - Affects color and makes the brick yellow.

5. Iron- < 7%
   - Gives red color to bricks (when excess of O_2 is available) and dark brown or even black color when insufficient O_2)
   - Improves impermeability and durability.
   - Tends to lower the fusion point of clay
   - Gives hardness and strength.
   - Excess of ferric oxide makes the brick dark blue.

Manufacturing of Clay Bricks

1. Preparation of brick earth

Unsoiling → Digging → Weathering → Blending → Tempering

↓

2. Moulding

↓

3. Drying

↓

4. Burning

↓

5. Brick

Manufacturing of Clay Bricks: Preparation of brick earth

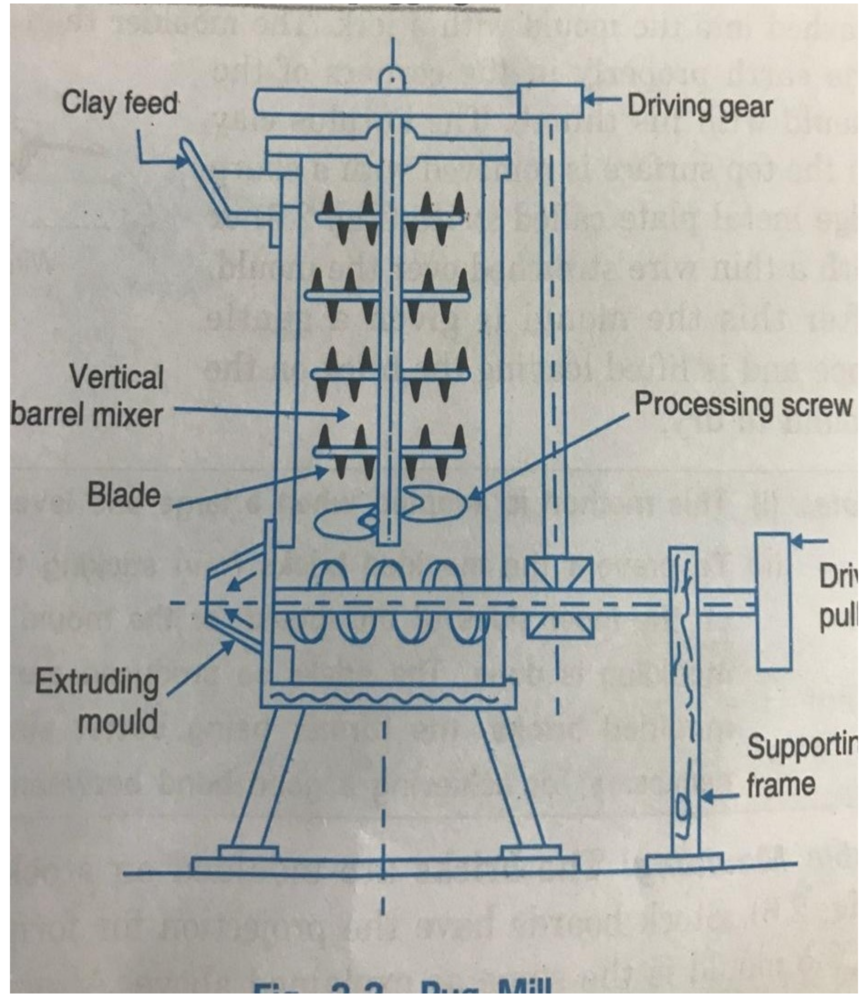
- Unsoiling
  - Making the soil free from gravels, coarse sand (> 2 mm), lime and kankar particles, organic matters.
  - Removal of about 20 cm of top layer of the earth.
- Digging
  - Soil mass is excavated , puddled, watered and left over for weathering and subsequent processing.
- Weathering
  - Soil heap is exposed to weathering to develop homogeneity in soil mass from different source and to eliminate the impurities by oxidation
  - Soluble solids in clay are also eroded due to rain.

Manufacturing of Clay Bricks: Preparation of brick earth

- Blending
  - Mixing of earth with sandy and calcareous earth in proportions to modify the soil composition
  - Additives such as fly ash, sandy loam, rice husk ash, basalt stone dust may be blended
  - Water is added in a controlled manner to obtain right consistency for moulding (easy mixing and workability), avoid excess moisture
- Tempering
  - Kneading of earth with feet so as to make the mass stiff and plastic. Plasticity refers to the property of wet clay to be permanently deformed without cracking
  - For manufacturing quality bricks, tempering is done in pug mills and operation is known as pugging.

Pug mill

- Conical iron tube sunk 60 cm in ground
- Blended earth along with required water is fed from top
- The knives cut through the clay and break lumps when the shaft rotates
- The yield from a pug mill is about 1500 bricks



Manufacturing of Clay Bricks (Contd.)

- Moulding
  - Process of giving shape to bricks from the prepared brick earth.
  - Methods of moulding

Moulding techniques	Water content in raw mix, %
Soft-mud or hand moulding i. Ground moulding ii. Table moulding	25-35
Extrusion and wire cutting ✓Soft extrusion ✓Stiff extrusion	 20-25 10-15
Semi-dry pressing	5-15
Dry pressing	<5

Manufacturing of Clay Bricks (Contd.)

- Drying
  - To remove the moisture so as to control shrinkage and saves fuel and time during burning.
  - Drying shrinkage depends on pore spaces within the clay and the amount of water mixed
  - Moisture content is brought down to about 3% under exposed conditions within 3-4 days. Thus, the strength of green bricks is increased and bricks can be handled safely
  - Bricks are open dried or artificially dried

- **Open driers**

Bricks are stacked on ground and dried in sunlight.

- **Artificial driers**

1. Hot floor drier- heat produced by furnace
2. Tunnel drier- heat produced by steam pipes, hot air pipes

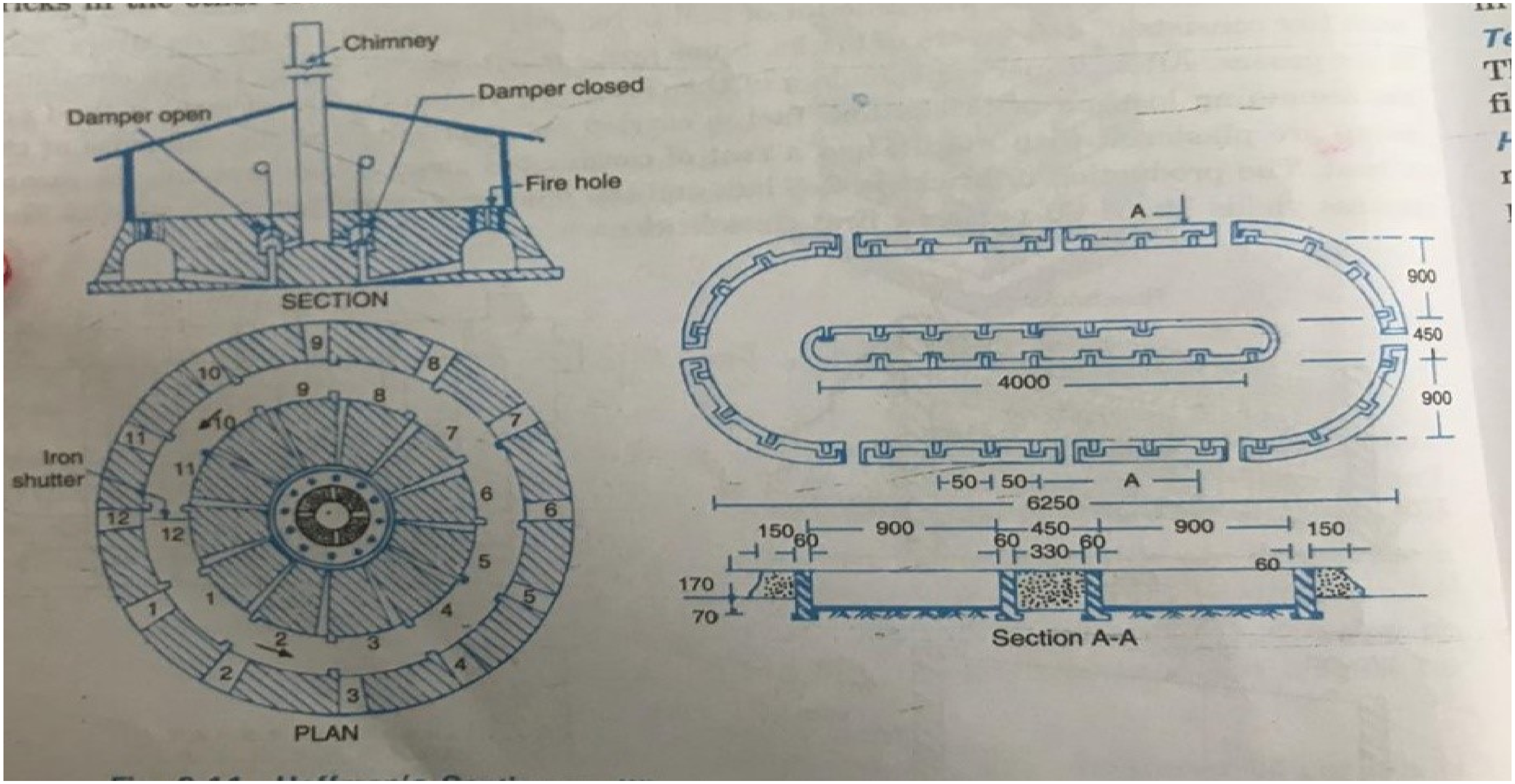
Manufacturing of Clay Bricks (Contd.)

- Burning-

  Burning of clay is done in kiln and may be divided into three stages

1. Dehydration(400-650^{o}C) water smoking stage
   - Water retained in pores of clay after drying is driven off and clay loses its plasticity.
2. Oxidation period (650-900^{o}C)
   - Remainders of carbons are eliminated, ferrous iron are oxidized to ferric form.
3. Vitrification (900-1100^{o}C)
   - To convert the mass into glass like substance.

Hoffman kiln



Classification of Bricks

- On basis of field practice

1. First class bricks-
   - ✓ Burnt thoroughly, deep cherry red in color.
   - ✓ Surface smooth and rectangular with parallel sharp and straight edges.
   - ✓ Free from flaws, cracks and stones and uniform texture.
   - ✓ A metallic and ringing sound should come when two bricks are struck
   - ✓ No impression should be left on brick when scratch is made by finger nail.
   - ✓ Water absorption should be 12-15% of its dry weight when immersed in cold water for 24 hrs.
   - ✓ Crushing strength of bricks should not be less than 10 N/mm^2 .
   - ✓ Uses-Pointing, exposed face work in masonry structure, flooring and reinforced brick works.

Classification of Bricks (contd.)

2. Second Class Bricks-

Supposed to have the same requirement as the first class ones except that-

- ✓ Small cracks and distortions are permitted.
- ✓ Water absorption of about 16-20%.
- ✓ The crushing strength should not be less than 7 N/mm^2 .
- ✓ Uses- All important or un important hidden masonry works and centering of reinforced bricks and RCC .

3. Third Class Bricks-

- ✓ Under-burnt.
- ✓ They are soft and light colored producing a dull sound and when struck against each other
- ✓ Water absorption about 25% of dry weight.
- ✓ Uses- building temporary structures.

Classification of Bricks (contd.)

4. Fourth Class Bricks-

- ✓ Over-burnt
- ✓ Badly distorted in shape and size and are brittle in nature.
- ✓ Uses- ballast of bricks in foundation and floors in concrete and road metals

Classification of Bricks (contd.)

- On the basis of strength

Bureau of Indian standard has classified the bricks on the basis of compression strength

Class	Avg Compressive Strength not less than (N/mm2)
35	35.0
30	30.0
25	25.0
20	20.0
17.5	17.5
15	15.0
12.5	12.5
10	10.0
7.5	7.5
5	5.0

Classification of Bricks (contd.)

- On basis of use
  - ✓ Common bricks - multi purpose manufactured economically
  - ✓ Facing bricks - used in front of building walls for pleasing appearance.
  - ✓ Engineering bricks - strong, impermeable used in all load bearing structure

- On the basis of finish
  - ✓ Sand facing - textured surface manufactured by sprinkling of sand on inner face.
  - ✓ Rustic brick - mechanically textured

Classification of Bricks (contd.)

- On the basis of burning
  - ✓ Pale brick- underburnt brick obtained from outer portion of kiln.
  - ✓ Body brick- well brunt brick obtained from central portion of kiln.
  - ✓ Arch brick- overburnt brick obtained from inner portion of brick.

- On basis of types
  - ✓ Solid bricks
  - ✓ Perforated bricks
  - ✓ Hollow bricks
  - ✓ Cellular bricks

Efflorescence Test on Bricks (IS:3495 Part-3)

1. Five samples are taken and the bricks are placed vertically in a dish 30 cm x 20 cm approximately in size with 2.5 cm immersed in distilled water.

2. The water is allowed to be absorbed by the brick and evaporated through it.

- NIL – there is no perceptible deposit of salt.
- Slight – Not more than 10% of the area of brick is covered with salt
- Moderate – heavy deposit covering upto 50% of the area of the brick
- Heavy – Heavy deposit covering more than 50% of the area of the brick

Water absorption Test on Bricks (IS:3495 Part-2)

1. Five bricks taken for test. They are allowed to dry in an oven at 105°C to 115°C till they attain a constant weight usually takes place in 48 hrs.

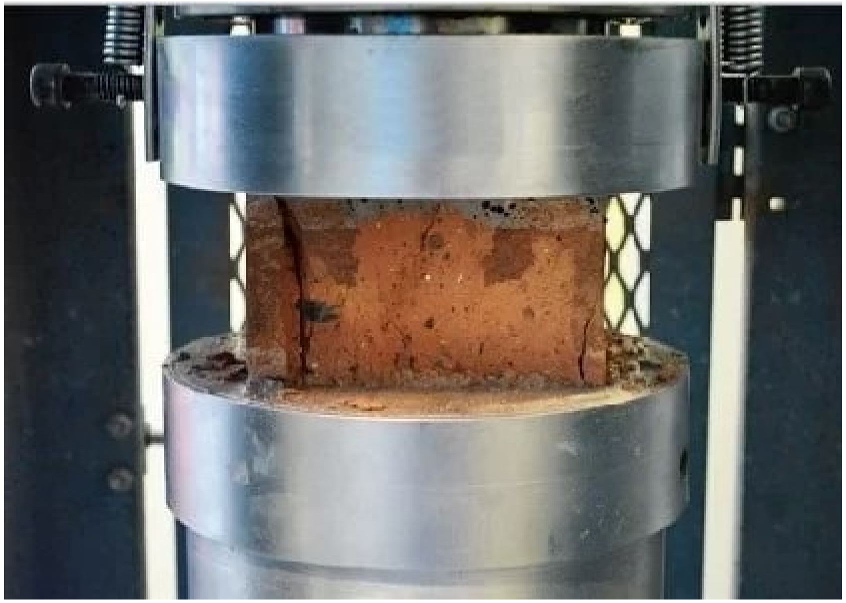
2. Then they are cooled at room temperature, which generally takes 4 to 6hrs without a fan and weight W1 is measured.

3. Then they kept in clear water at 27±2°C for 24 hrs and wiped dry with a damp cloth and weight W2 Is measured.

4. The average percentage of water absorbed as percentage of dry weight is reported.

Compressive Strength Test on Bricks (IS:3495 Part-1)

1. Five bricks taken at random and their dimensions are measured
2. Immersed in water of 27±2°C for 24 hrs.
3. Surplus moisture is allowed to drain and the frog, if any, is filled with mortar 1:3 (1 part cement, 3 parts clean coarse sand 3 mm down).
4. Bricks are stored under a jute bag for another 24hrs after which they are immersed in clean water for 3 days for curing.
5. At the time of testing, these bricks are removed from water, wiped dry and placed with the flat surface between plywood sheets each of 3 mm thickness.
6. This specimen is kept under UTM and load is applied 14 MPa/minute till failure
7. The compressive strength is noted down

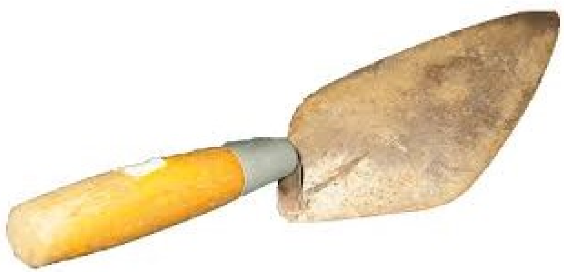


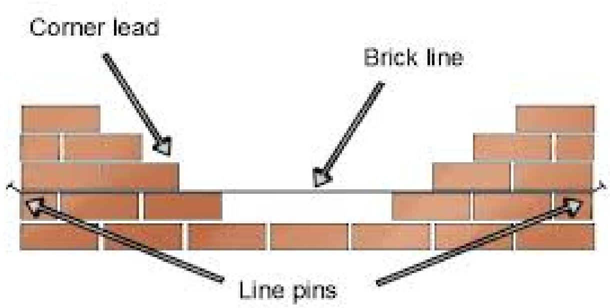
Dimensional tolerance

1. Bricks are selected at random to check measurement of length, width, height.
2. Variations in dimensions are allowed only within narrow limits, ± 3% for class one and ± 8% for other classes.

Brick Masonry

- Types of bonds in brick masonry wall construction are classified based on laying and bonding style of bricks in walls.
- The bonds in brick masonry is developed by filling the mortar

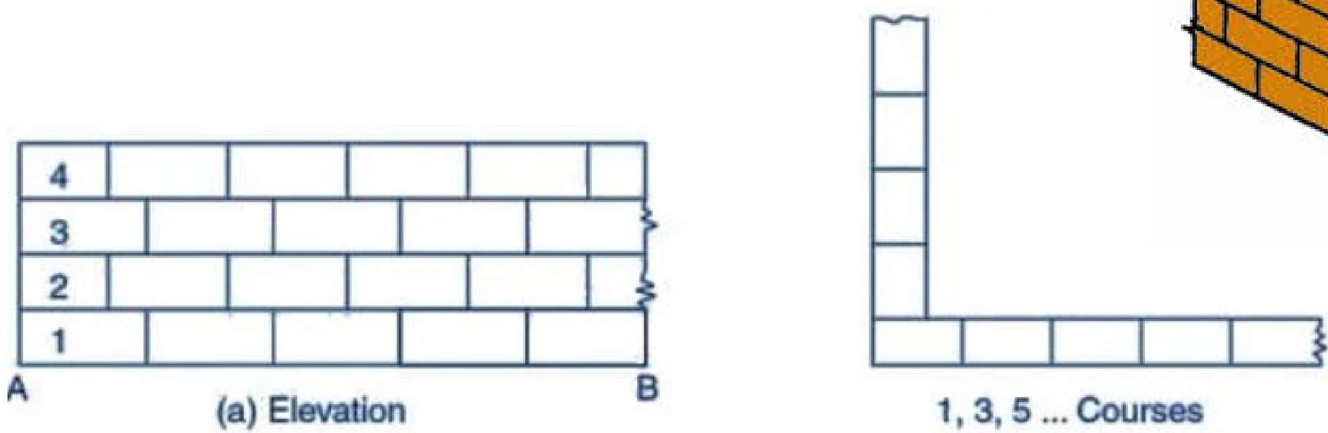




Types of bonds

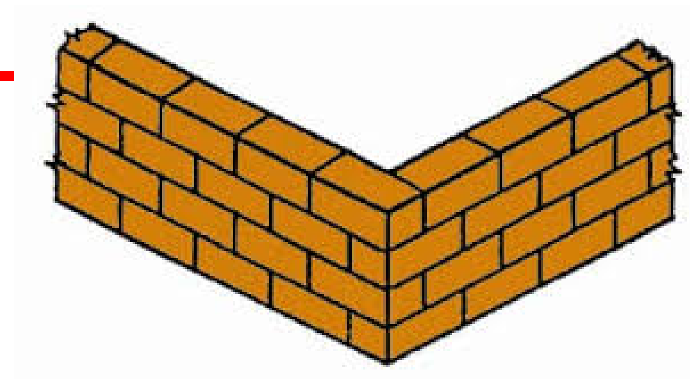
- **The most commonly used types of bonds in brick masonry are:**
  - Stretcher bond
  - Header bond
  - English bond and
  - Flemish bond
- **Other Types of bonds are:**
  - Facing bond
  - Dutch bond
  - English cross bond
  - Brick on edge bond
  - Raking bond
  - Zigzag bond
  - Garden wall bond

Stretcher bond



(a) Elevation

(b) Plan



- It is also called as running bond
- Stretcher bonds are commonly used in the steel or reinforced concrete framed structures as the outer facing.
- They are not stable for longer span and height and thus they require supporting structures such as brick masonry columns at regular intervals

Header bond

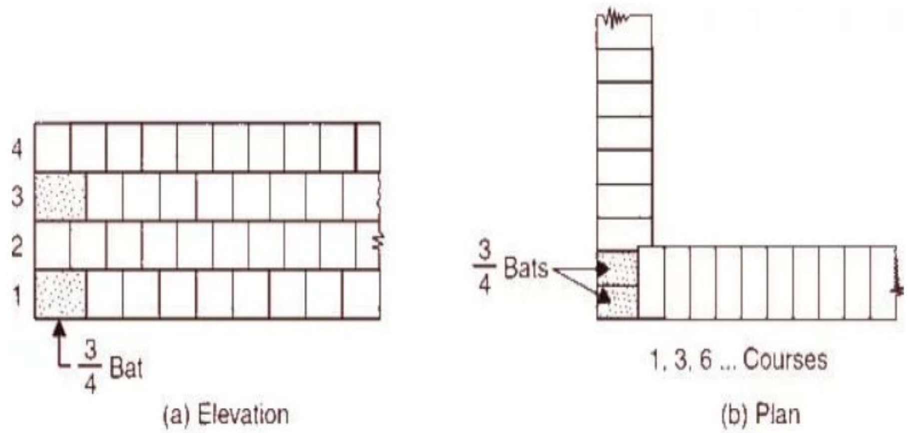
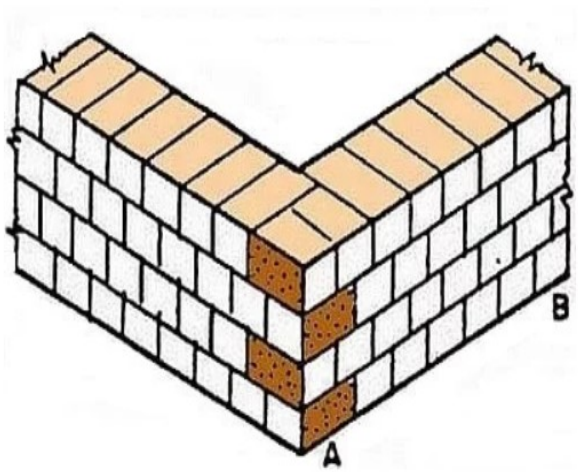


Fig-2: Header Bond



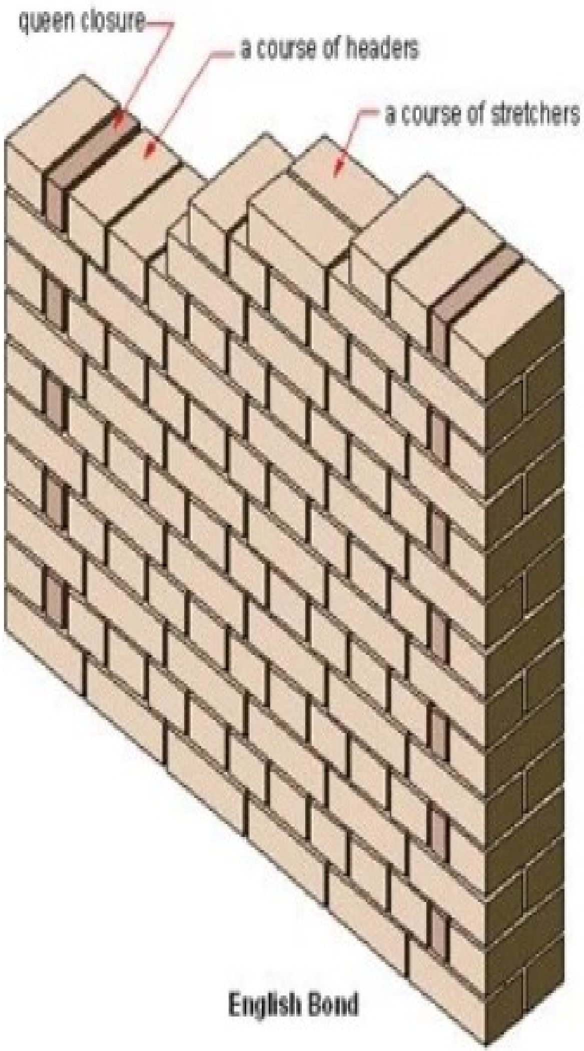
- Header is the shorter square face of the brick
- In header bonds, all bricks in each course are placed as headers on the faces of the walls.
- In header bonds, the overlap is kept equal to half width of the brick. To achieve this, **three quarter brick bats** are used in alternate courses as quoins.

Quoin

Quoins are large rectangular blocks of masonry or brick that are built into the corners of a wall.

They can be used as a load-bearing feature to provide strength and weather protection, but also for aesthetic purposes to add detail and accentuate the outside corners of a building.

English Bond



- English bond in brick masonry has one course of stretcher only and a course of header above it, i.e. it has two alternating courses of stretchers and headers.
- To break the continuity of vertical joints, quoin closer is used in the beginning and end of a wall after first header.
- A quoin close is a brick cut lengthwise into two halves and used at corners in brick walls.

https://www.youtube.com/watch?v=2zjXPrnMVpk

Flemish bond

- Flemish bond, also known as Dutch bond, is created by laying alternate headers and stretchers in a single course.
- The next course of brick is laid such that header lies in the middle of the stretcher in the course below, i.e. the alternate headers of each course are centered on the stretcher of course below.
- Every alternate course of Flemish bond starts with header at the corner.